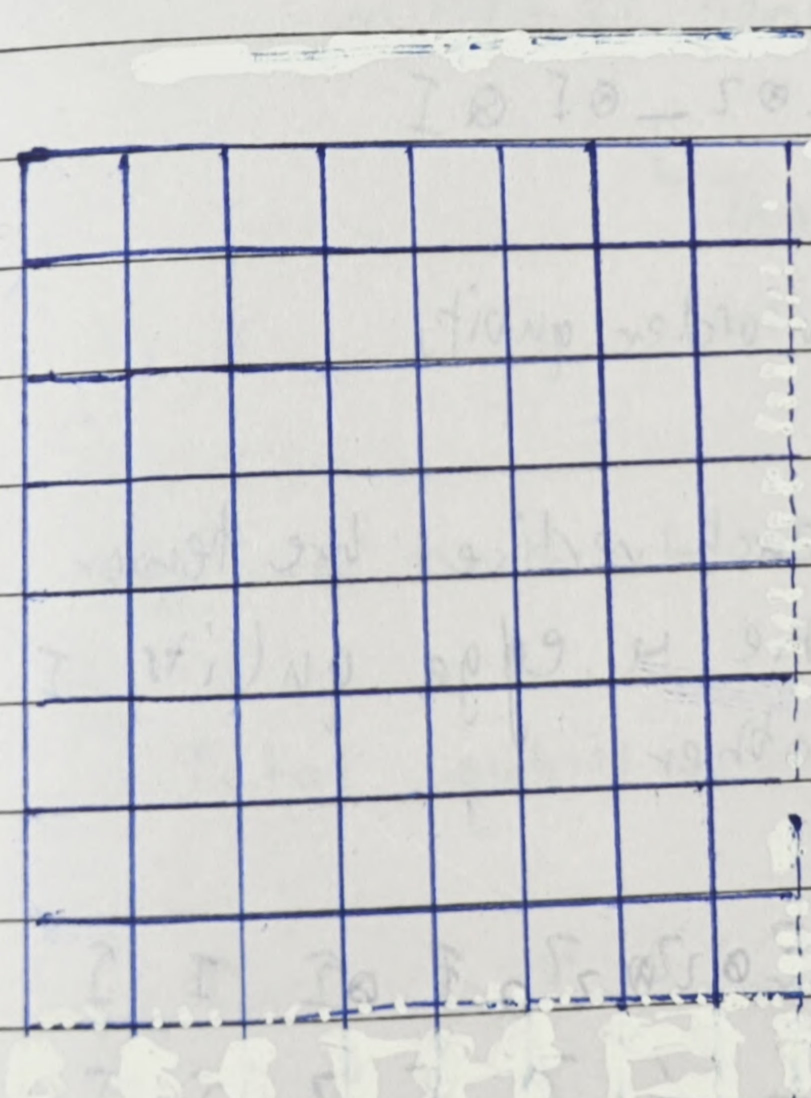
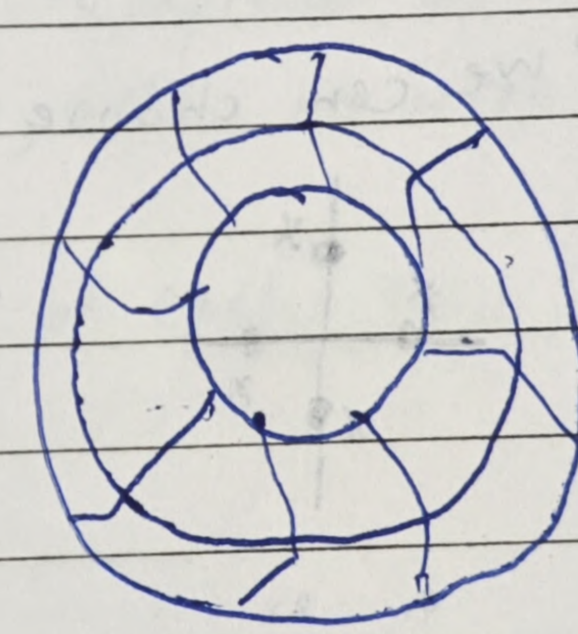
* Toric Code

- A CSS code
- It's a family of codes, one for each positive integer ≥ 2
- All stabilizer generators have weight 4
- It has geometric locality — each stabilizer generator measures qubits only near it
- It has increasingly large distance, can tolerate relatively high error rate
- Toric Code Description:

Consider $L \times L$ lattice for $L=8$ — $L \geq 2$



→

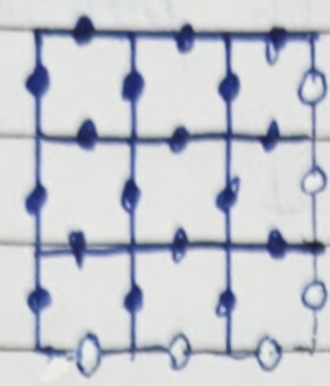


Torus / Doughnut

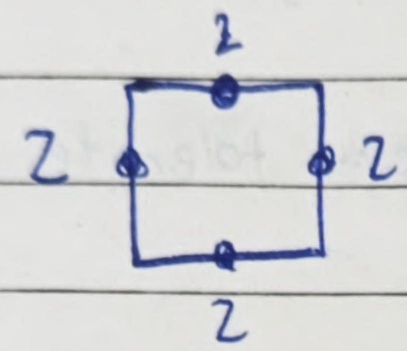
↳ Bottom dotted line is the same as top line,
Right dotted line is the same as left line

- Qubits are placed on the edges, indication by solid circles

↳ $2L^2$ qubits



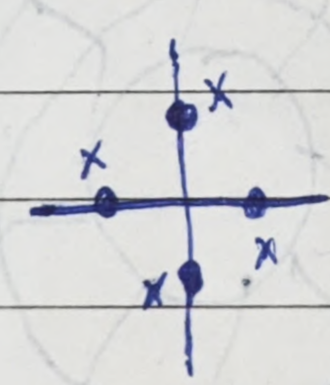
- we obtain stabilizer generators by,



For each tile we tensor Z
on the 4 edge qubits and
I on all other

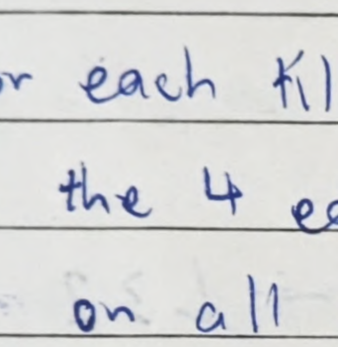
$I \otimes I \_\_ \otimes \overset{1}{Z} \otimes Z \otimes Z \otimes Z \otimes I \otimes I$

↳ we can choose how to order qubits



- For each vertice we tensor X on the 4 edge qubits, I on all other

- Ex, →



→ Z Z Z Z I I I I

I Z Z Z I Z I I

Z I I I Z I Z Z

I I X I X X X I   I Z I I I Z Z Z

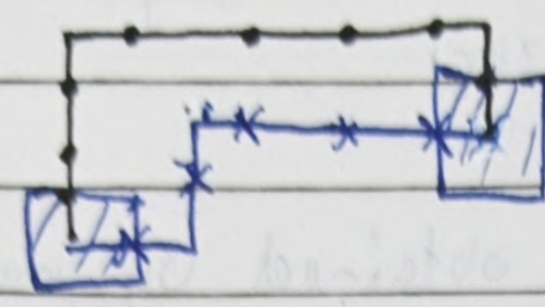
I I I X X X I X   X X X I I I I X

I I X I X X I X   X X I X I I I X

X X X I I I X I   X X X I I I X I

* Correcting Errors

when we get the -1 outcomes, we apply X correction gates along the shortest path



→ Forms closed loop.

↓

Errors that go around the torus aren't corrected

— → Actual error path

— → Shortest Path

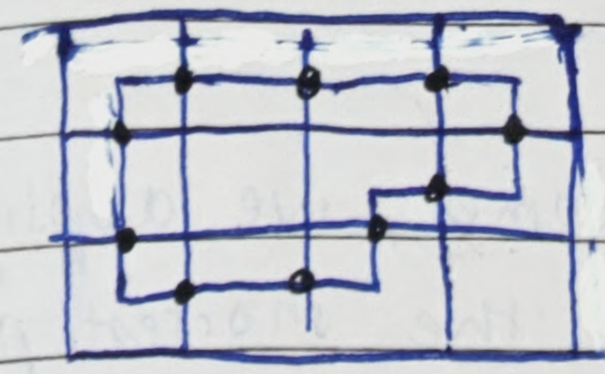
X → Qubit corrected by X gate

- For more than 2 -1 outcomes, we can we can use Blossom algorithm (classical algorith) to find minimum-weight perfeet matching paths

→ It works well below a noise rate of ~10%

↓

Detecting, correcting Z errors is analogaus with X errors, we use X-sg instead of Z-sg



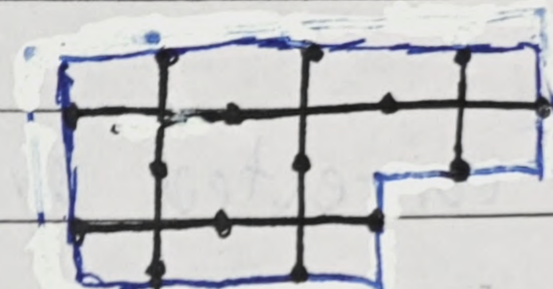
→ When error forms closed loop, we can't detect it

↳ However, it isn't actually an error as it is in the stabilizer.

↓

It can be obtained by multiplying adjacent X-sg forming a closed loop

↓



→ Closed loops of this form, that go around the torus, can't be detected

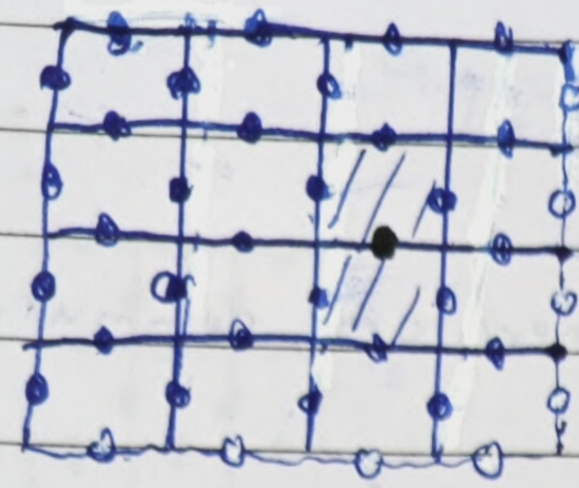
→ Min weight → L → distance

Toric Code → $[[2L^2, 2, L]]$ code

↓

Topological Quantum Error Correcting Code

* Detecting errors



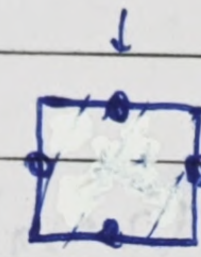
• → unaffected qubit

• → qubit affected by X error

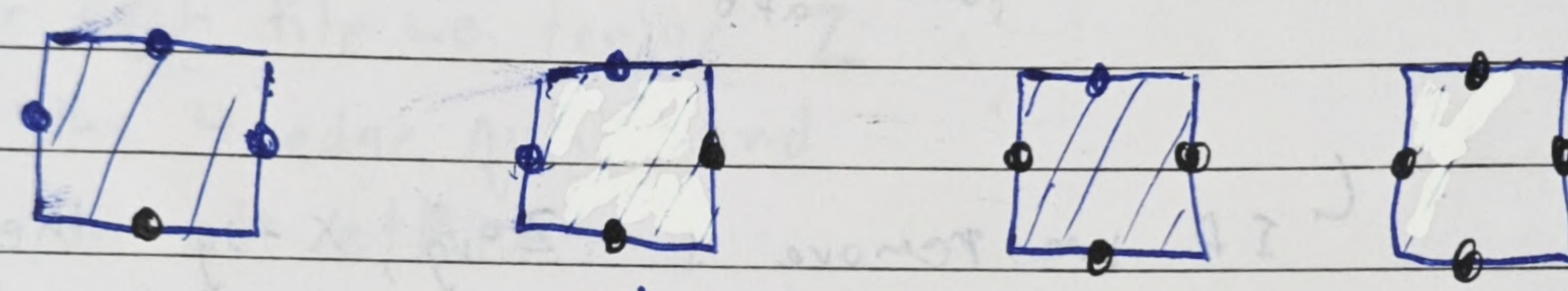
□ → +1 measurement outcome

▨ → -1 measurement outcome

X-error → corrected by Z-sq



+1

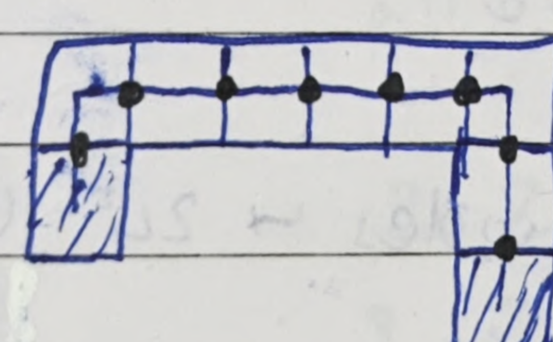


-1 +1 -1 +1

↳ Z-sq measures parity of 4 qubits

↳ +1 doesn't mean no error always

In case of multiple X errors,



unaffected qubits aren't shown

↳ Code gives -1 outcomes for Z-sq at endpoints of the chain

↳ It can be detected, corrected

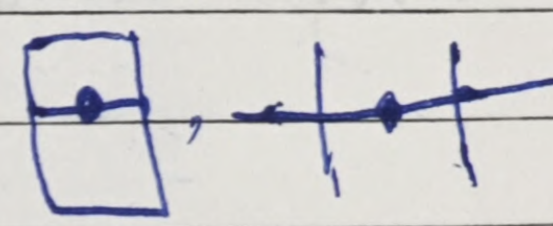
XXIXIIIX

~~X~~XXIIIXI

- we can see, all stabilizer generators commute each other

$$\prod (Z\text{-}sg) \cdot \prod (X\text{-}sg) = I$$

↳ All Z, X repeat 2 times in 2 squares

↳ 

$\therefore (Z\text{-}sg)_b = \prod_{a \neq b} (Z\text{-}sg)_a \quad , (X\text{-}sg)_b = \prod_{a \neq b} (X\text{-}sg)_a$

↳ If we remove 1 Z-sg, X-sg they form minimal generating set

↳ That 1 can be expressed as the product of all others, adds no extra information

so, we have L^2-1 sg of each type, $2L^2-2$ in total

Total qubits → $2L^2$

Toric code encodes → $2L^2-(2L^2-2) = 2$ qubits